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C204

Midterm Lab Task 3 - Python List Collections

Problem 1. Using List Collection type. Create a program that will allow the user to perform the following **functions**: (add, update, search, delete, display, and sort) items in a list:

Code:

```
1 def main():
2     # List to store items
3     items = []
4
5     while True:
6         print("\n[ MENU OPTIONS ]")
7         print("1 - Add Items")
8         print("2 - Search for an Item")
9         print("3 - Remove an Item")
10        print("4 - View all Items (Sorted A-Z | Z-A)")
11        print("0 - Exit Program")
12
13        choice = input("Pick one [0 to quit]: ")
14
15        # Exit program
16        if choice == "0":
17            print("Exiting program... Goodbye!")
18            break
19
20        # Add items
21        elif choice == "1":
22            print("\nEnter items (type 'x' to stop):")
23            while True:
24                item = input("Enter item: ")
25                if item.lower() == "x":
26                    break
27                items.append(item)
28            print("Items added successfully!")
29
30        # Search for item
31        elif choice == "2":
32            search_item = input("Enter item to search: ")
33            count = items.count(search_item)
34            if count > 0:
35                print(f"'{search_item}' found {count} time(s) in the list.")
36            else:
37                print(f"'{search_item}' not found in the list.")
38
```

```

39         # Remove item
40         elif choice == "3":
41             remove_item = input("Enter item to remove: ")
42             if remove_item in items:
43                 items.remove(remove_item)
44                 print("Item found and deleted.")
45             else:
46                 print("Item not found - deletion unsuccessful.")
47
48         # View items sorted
49         elif choice == "4":
50             if len(items) == 0:
51                 print("The list is empty.")
52             else:
53                 sort_choice = input("Sort order (A for Ascending, D for Descending): ").lower()
54                 if sort_choice == "a":
55                     for i in sorted(items):
56                         print(i)
57                 elif sort_choice == "d":
58                     for i in sorted(items, reverse=True):
59                         print(i)
60                 else:
61                     print("Invalid choice. Showing unsorted list:")
62                     for i in items:
63                         print(i)
64
65         else:
66             print("Invalid choice! Please try again.")
67
68 # Run the program
69 main()

```

Sample output:

```

[ MENU OPTIONS ]
1 - Add Items
2 - Search for an Item
3 - Remove an Item
4 - View all Items (Sorted A-Z | Z-A)
0 - Exit Program
Pick one [0 to quit]: 1

Enter items (type 'x' to stop):
Enter item: A
Enter item: B
Enter item: C
Enter item: D
Enter item: E
Enter item: x
Items added successfully!

[ MENU OPTIONS ]
1 - Add Items
2 - Search for an Item
3 - Remove an Item
4 - View all Items (Sorted A-Z | Z-A)
0 - Exit Program
Pick one [0 to quit]: 2
Enter item to search: C
'C' found 1 time(s) in the list.

```

```
[ MENU OPTIONS ]
1 - Add Items
2 - Search for an Item
3 - Remove an Item
4 - View all Items (Sorted A-Z | Z-A)
0 - Exit Program
Pick one [0 to quit]: 4
Sort order (A for Ascending, D for Descending): A
A
B
C
D
E
```

```
[ MENU OPTIONS ]
1 - Add Items
2 - Search for an Item
3 - Remove an Item
4 - View all Items (Sorted A-Z | Z-A)
0 - Exit Program
Pick one [0 to quit]: 3
Enter item to remove: B
Item found and deleted.
```

```
[ MENU OPTIONS ]
1 - Add Items
2 - Search for an Item
3 - Remove an Item
4 - View all Items (Sorted A-Z | Z-A)
0 - Exit Program
Pick one [0 to quit]: 0
Exiting program... Goodbye!
```

```
• === Code Execution Successful ===
```